passes through the intersection of the two given planes whatever be the value of k.

It will be perpendicular to $4x + 5y - 3z + 1 = 0$

$\therefore\ 2.4 - (1 + k)\,5 + 3k(-3) = 0$ i.e. $14k = 3$ or, $k = 3/14$

Thus the required plane is $2x - y + 3(3z - y)/14 = 0$

or, $28x - 17y + 9z = 0$

Ex. 8. Find the equation of the plane through the pts. (2, 2, 1) and (9, 3, 6) and perpendicular to the plane $2x + 6y + 6z = 9$

[C.H. 1980; C.U. 1981; R.U. 1962, '78; D.H. 1976] D.U.(P) 1986]

Any plane through (2, 2, 1) is

$a(x - 2) + b(y - 2) + c(z - 1) = 0$ ... ... (1)

Since it passes through (9, 3, 6) then (1) becomes

$a(9 - 2) + b(3 - 2) + c(6 - 1) = 0$ or, $7a + b + 5c = 0$ ... ... (2)

The plane (1) is perpendicular to $2x + 6x + 6z - 9 = 0$

Hence $2a + 6b + 6c = 0$ ... ... (3)

From (2) and (3) by cross multiplication.

$$\frac{a}{-24} = \frac{b}{-32} = \frac{c}{40} \text{ or, } \frac{a}{3} = \frac{b}{4} = \frac{c}{-5} \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

Putting the values of a, b, c, from (4) in (1), the equation of the required plane is $3(x - 2) + 4(y - 2) - 5(z - 1) = 0$ or, $3x + 4y - 5z = 9$

Alternate Method : The required plane is

$$\begin{vmatrix} x-2 & y-2 & z-1 \\ 9-2 & 3-2 & 6-1 \\ 2 & 6 & 6 \end{vmatrix} = 0$$

or, $3x + 4y - 5z = 9$ by Art. (11)

Ex. 9. Show that the equation to the plane through P (2, 3, — 1) at right angles to OP is $2x + 3y - z = 14$

The direction cosines of the line OP are $\frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, \frac{-1}{\sqrt{14}}$

Any plane at right angles to OP can be written as

$$\frac{2}{\sqrt{14}}x + \frac{3}{\sqrt{14}}y - \frac{1}{\sqrt{14}}z = p$$

Where $p = \sqrt{2^2 + 3^2 + (-1)^2} = \sqrt{14}$)

The required plane is $2x + 3y - z = 14$

Ex. 10. A plane meets the Co-ordinate axes in A, B, C such that the centroid of the triangle ABC is the point (p, q, r); show that the equation of the plane is [C.U. 1982]

$$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 3$$

Let the equation of the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ ... ... (1)

$\therefore$ A is (a, o, o) B is (o, b, o) and C is (o, o, c)

Since the plane parallel to x – axis.

So, normal to the plane and x–axis are perpendicular.

$\therefore 1.a + 0.b + 0.c = 0$

From (2); $a - 4b + c = 0$

By the rule of cross multiplication

$\therefore \frac{a}{0-0} = \frac{b}{0-1} = \frac{c}{-4-0}$

$\therefore \frac{a}{0} = \frac{b}{-1} = \frac{c}{-4} = k$ (say)

$\therefore \quad a = 0K, b = -K, C = -4K$

i,e, $a = 0, b = -1, c = -4$

Putting the values of a, b, c in (i) we get

$0(x-2) + 1(y-3) + 4(z-1) = 0$

$y + 4z - 7 = 0$

Which is the required equation.

17. Find the equation of the plane perpendicular to the planes $x - 4y + z = 0$ and $3x + 4y + z = 2$ and at a unit distance from the origin. [N.U.H 1997]

Solution : The given planes

$x - 4y + z = 0$ ... ... (1)

and $3x + 4y + z = 2$ ... ... (2)

Let equation of the plane which perpendicular both planes (1) and (2) is

$ax + by + cz + d = 0$ ... ... (3)

Since the plane (3) perpendicular both the planes (1) and (2)

$\therefore a - 4b + c = 0$

and $3a + 4b + c = 0$

$\therefore \frac{a}{-4-4} = \frac{b}{3-1} = \frac{c}{4+12}$

$\frac{a}{-8} = \frac{b}{2} = \frac{c}{16} = k$ (say)

$\therefore a = -8k, b = 2k, c = 16k$

$a = -4, b = 1, c = 8$

Pulting the values of a, b, c in (3) we get (3) $\Rightarrow -4x + y + 8z + d = 0$ ... ... (4)

According to the problem,

$\frac{-4.0 + 0 + 8.0 + d}{\sqrt{4^2 + 1^2 + 8^2}} = \pm 1$

$\left|\frac{d}{9}\right| = \pm 1 \therefore d = \pm 9$

Otherwise we shall find the equation of the plane passing through any three points say $(0, -1, -1)$, $(4, 5, 1)$ and $(3, 9, 4)$ and then show that the fourth point $(-4, 4, 4)$ satisfied the plane.

Equations of any plane through $(0, -1, -1)$, $(4, 5, 1)$ and $(3, 9, 4)$ is

$5x - 7y + 11z + 4 = 0$ by example (1)

The point $(-4, 4, 4)$ clearly satisfies this plane. Hence the four points are coplanar.

N.B : For finding out the equation to the plane use, Cor. 3

Ex. 3. Find the equation of the plane passing through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $4x + 3y + 2z + 1 = 0$ and the point $(1, 2, 3)$

Any plane through the intersection of the two planes is

$x + 2y + 3z + 4 + k(4x + 3y + 2z + 1) = 0$ ... ... (1)

Since it passes through $(1, 2, 3)$,

$18 + 17k = 0$ or $k = -18/17$

Putting the value of k in (1) we have

$$x + 2y + 3z + 4 - \left(\frac{18}{17}\right)(4x + 3y + 2z + 1) = 0$$

or, $55x + 20y - 15z - 50 = 0$ or, $11x + 4y - 3z = 10$

which is the required plane.

Ex. 4. Find the angle between the planes $2x - y + z = 6$ and $x + y + 2z = 7$

Let θ be the angle between the planes then

$$\cos\theta = \frac{2.1 + (-1).1 + 1.2}{\sqrt{\{2^2 + (-1^2) + 1^2\}}\ \sqrt{(1^2 + 1^2 + 2^2)}} = \frac{3}{\sqrt{6}\sqrt{6}} = \frac{1}{2} \quad \therefore \theta = \frac{\pi}{3}$$

Ex. 5. Find the equation of the plane through the point $(4, -2, 1)$ and parallal to the plane whose direction numbers are $7, 2, -3$.

The equation to the plane through $(4, -2, 1)$ is $a(x - 4) + b(y + 2) + c(z - 1) = 0$. The condition that the co-efficients a, b, c are proportional to the direction numbers $7, 2, -3$, therefore, the equation of the required plane is $7k(x - 4) + 2k(y + 2) - 3k(z - 1) = 0$ where k is constant.

or, $7(x - 4) + 2(y + 3) - 3(z - 1) = 0$ or, $7x + 2y - 3z - 21 = 0$

Ex. 6. Find the equation of the plane through the point $(4, 0, 1)$ and parallel to the plane $4x + 3y - 12z + 6 = 0$

The equations of the parallel planes differ by a constant only. Hence the equation of the plane parallel to $2x + 3y - 12z + 6 = 0$ is $4x + 4y - 12x + k = 0$ Since it passes through $(4, 0, 1)$.

$16 + 0 - 12 + k = 0$. or, $k = -4$

The required plane is $4x - 3y - 12z - 4 = 0$

Ex. 7. Find the equation of the plane passing through the lines of intersection of the planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to the plane $4x + 5y - 3z + 1 = 0$

The plane $2x - y + k(3z - y) = 0$

i.e. $2x - (1 + k)y + 3kz = 0$

18. Find the equation of the plane containing the point (1, –1, 1) and passing through the point of intersection of the planes $x - 2y + 3z + 4 = 0$ and $2x - 3y + 4z - 7 = 9$. [N.U.H-2001]

Solution : Given planes

$x - 2y + 3z + 4 = 0$ ... ... (1)

$2x - 3y + 4z - 7 = 0$ ... ... (2)

Let equation of the plane through the line of intersection of the planes (1) and (2) is $x - 2y + 3z + 4 + k(3x - 3y + 4z - 7) = 0$ ... ... (3)

Where k is an arbitrary constant.

It passes through the point (1, –1, 1).

$\therefore 1 - 2.(-1) + 3.(1) + 4 + k\{2.1 - 3(-1) + 4.1 - 7\} = 0$

$1 + 2 + 3 + 4 + k(2 + 3 + 4 - 6) = 0$

$10 + 2k = 0$

$\therefore k = -5$

Putting the value of k in (3) we get

$x - 2y + 3z + 4 - 5(2x - 3y + 4z - 7) = 0$

$\Rightarrow x - 2y + 3z + 4 - 10x + 15y - 20z + 35 = 0$

$\Rightarrow -9x + 13y - 17z + 39 = 0$

$\Rightarrow 9x - 13y + 17z - 39 = 0$

Which is the required plane.

19. Find the equatin of the plane passing through the line of intersection of the planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to the plane $4x + 5y - 3z + 7 = 0$. [N.U.H – 2004]

Solution : Given planes

$2x - y = 0$ ... ... (1)

$3z - y = 0$ ... ... (2)

$4x + 5y - 3z + 7 = 0$ ... ... (3)

Let equation of the plane through the line of intersection of the planes (1) and (2) is

$2x - y + k(3z - y) = 0$ ... ... (4)

Where k is an arbitrary Constant

$\Rightarrow 2x - (1 + k)y + 3kz = 0$

Since the plane (3) and (4) are perpendicular

$a_1a_2 + b_1b_2 + c_1c_2 = 0$

$\therefore 4.2 + 5.\{-(1 + k)\} - 3.3k = 0$

$\Rightarrow 8 - 5 - 5k - 9k = 0$

$\Rightarrow 3 = 14k \therefore k = \frac{3}{14}$

Putting the values of k in (4) we get

$2x - y + \frac{3}{14}(3z - y) = 0$

$\Rightarrow 28x - 14y + 9z - 3y = 0$

$\Rightarrow 28x - 17y + 9z = 0$

Which is the required plane.

20. A point p moves on a fixed plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. the plane through p perpendicular to op meets the axes in A, B, C respectively. If the planes are drawn through A, B, C parallel to the co-ordinate planes then show that the locus of the point inter section is

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz}.$$

Solution : The given plane

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \ldots \quad \ldots \quad (1)$$

Let the point $P(\alpha, \beta, \gamma)$. passes through (1)

$$\therefore \frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = 1 \quad \ldots \quad \ldots \quad (2)$$

Let equation of the plane through $P(\alpha, \beta, \gamma)$ perpendicular to op is

$\alpha(x - \alpha) + \beta(y - \beta) + \gamma(z - \gamma) = 0$

$$\Rightarrow \frac{\alpha x + \beta y + \gamma z}{\alpha^2 + \beta^2 + \gamma^2} = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha} + \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta} + \frac{z}{\alpha^2 + \beta^2 + \gamma^2} = 1 \quad \ldots \quad \ldots \quad (3)$$

Since the plane (3) meets the axes in A, B, C.

$\therefore$ Co-ordinates of A $\left(\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, 0, 0\right)$

---------- B $\left(0, \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, 0\right)$

---------- C $\left(0, 0, \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma}\right)$

The planes through A, B, C Parallel to the Co-ordinate planes intersect at Q (say)

$\therefore$ Co-ordinate of Q $\left(\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma}\right)$

i.e. at the point Q

$$x = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, y = \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, z = \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma}$$

$$\text{Now, L.H.S} = \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \left(\frac{\alpha}{\alpha^2 + \beta^2 + \gamma^2}\right)^2 + \left(\frac{\beta}{\alpha^2 + \beta^2 + \gamma^2}\right)^2 + \left(\frac{\gamma}{\alpha^2 + \beta^2 + \gamma^2}\right)^2$$

$$= \frac{\alpha^2 + \beta^2 + \gamma^2}{(\alpha^2 + \beta^2 + \gamma^2)^2}$$

$$\therefore \frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{\alpha^2+\beta^2+\gamma^2}$$

$$\text{R.H.S} = \frac{1}{ax}+\frac{1}{by}+\frac{1}{cz} = \frac{\alpha}{a(\alpha^2+\beta^2+\gamma^2)}+\frac{\beta}{b(\alpha^2+\beta^2+\gamma^2)}+\frac{\gamma}{c(\alpha^2+\beta^2+\gamma^2)}$$

$$= \frac{\alpha/a+\beta/b+\gamma/c}{\alpha^2+\beta^2+\gamma^2}$$

$$\therefore \frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}=\frac{1}{\alpha^2+\beta^2+\gamma^2}, \text{ [By --(2)]}$$

$\therefore$ R.H.S = L.H. S (Proved)

21. Find the equation of the plane through the points (1, 0, –1) and (2, 1, 3) and perpendicular to the plane $2x + y + z = 1$. [N.U.H -2005]

Solutions : The given plane

$$2x + y + z = 1 \quad \ldots \quad \ldots \quad (1)$$

Let equation of the plane through the point (1, 0, – 1) is

$$a(x-1)+b(y-0)+c(z+1)=0 \quad \ldots \quad \ldots \quad (2)$$

It passes through (2, 1, 3).

$$\therefore a(2-1)+b(1-0)+c(3+1)=0$$

$$\Rightarrow a+b+4c=0 \quad \ldots \quad \ldots \quad (3)$$

Sinec the (1) and (2) ore perpendicular.

$$\text{So, } 2a+b+c=0 \quad \ldots \quad \ldots \quad (4)$$

From (3) and (4), we get by the rule of cross multiplication

$$\frac{a}{1-4}=\frac{b}{8-1}=\frac{c}{1-2}$$

$$\frac{a}{3}=\frac{b}{7}=\frac{c}{-1}=k \text{ (say)}$$

$\therefore a = -3k, b = 7k, c = -k$

Putting the values of a, b, c in (2) we get

$$(2) \Rightarrow -3k(x-1)+7k(y-0)-k(z+1)=0$$

$3x - 7y + z - 2 = 0$ Which is required plane.

22. A variable plane passes through a fixed point (a, b, c) and meets the axes in A, B, C. Show that the locus of the point of intersection of the planes through A, B, C parallel to the Co-ordinate planes is $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=1$ [N.U.H-2006]

Solution : Let the variable plane be

$$\frac{x}{a_1}+\frac{y}{b_1}+\frac{z}{c_1}=1 \quad \ldots \quad \ldots \quad (1)$$

it passes through the point (a, b, c)

$$\therefore \frac{a}{a_1}+\frac{b}{b_1}+\frac{c}{c_1}=1 \quad \ldots \quad \ldots \quad (2)$$

Again, plane (1) meets the axes in A, B, C.

$\therefore$ Co-ordinates of A $(a_1, 0, 0)$

---------- B $(0, b_1, 0)$

---------- C $(0, 0, c_1)$

Planes through A, B, C parallel to the Co-ordinate planes intersect at M (say)

$\therefore$ Co-ordinate of M (a_1, b_1, c_1)

At the point M; $a_1 = x, b_1 = y, c_1 = z$.

Putting the values of a_1, b_1, c_1 in (2) we get $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=1$. which is the required locus.

23. Two system of rectangular axes have the same origin. If a plane cuts the axes at a, b, c and $a_1, b_1 c_1$ respectively from the origin. Prove that $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c_1^2}$

Sol : There are two rectangular system.

Eq. of the plane w. r. to the system (1) and (2) and are

$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ ... ... (1)

and $\frac{x}{a_1}+\frac{y}{b_1}+\frac{z}{c_1}=1$ ... ... (2)

Perpendicular distanec from the origin to planes (1) and (2) are

$$d_1=\frac{0+0+0-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}$$

$$d_2=\frac{0+0+0-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=\frac{1}{\sqrt{\frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c_1^2}}}$$

Since Two system have same origin

$$\text{So } d_1 = d_2 \Rightarrow \frac{0+0+0-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=\frac{1}{\sqrt{\frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c_1^2}}}$$

$=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c_1^2}$ (Proved)

24. Find the equation of the plane passing through the points (1, 2, –2) and (3, –2, 6) and perpendicular to the plane $2x - y - z + 7 = 0$ [N.U. H-2009]

Solution : The Given plane

$2x - y - z + 7 = 0$ .... (1)

Let equation of the plane through the point (1, 2, –2) is

$a(x - 1) + b(y - 2) + c(z + 2) = 0$ ... ... (2)

It passes through (3, – 2, 6)

$\therefore a(3-1)\,b(-2-2)+c(6+2)=0$

$\Rightarrow 2a-4b+8c=0$ ... (3)

$\Rightarrow a-2b+4c=0$ ... ...

According the problem, planes (1) and (2) are perpendicular.

$\therefore 2a-b-c=0$ ... ... (4)

From (3) and (4), by the rule of cross multiplication

$$\frac{a}{2+4}=\frac{b}{8+1}=\frac{c}{-4+1}$$

$$\frac{a}{6}=\frac{b}{9}=\frac{c}{-3}=k \text{ (say)}$$

$\therefore a=6k, b=9k, c=-3k$

Putting the values of a, b, c in (2) we get, $6x(x-1)+9k(y-2)-3k(z+2)=0$

$2x+3y-z-10=0$

Which is required equation of the plane

EXERCISE II

1. Find the general equations of planes perpendicular to

(1) x-axis (2) y-axis (3) xoy plane (4) xoz plane.

Ans. (1) $x=p$; (2) $y=p$; (3) $lx+my=p$; (4) $lx+nz=p$

2. Find the equation of the plane through the points.

(a) (2, 1, 3). (−1, −2, 4), (4, 2, 1) Ans. $5x-4y+3z-15=0$

(b) (1, 1, −1) (−2, −2, 2), (−1, −1, 2) Ans. $x-3y-2z=0$

(c) (2, 1, −3). (3, −1, 4) (7, 5, 6) Ans. $23z-13y-7x-54=0$

3. Show that the four points are coplanar

(a) (0, 4, 3). (−1, −5, −3). (−2, −2, 1). (1, 1, −1)

(b) (−6, 3, 2), (3, −2, 4), (5, 7, 3) and (−13, 17, −1)

4. Show that the line of intersection of the first two planes is co-planar with the line of intersection of the latter two and find the equation of the plane containing two lines.

(a) $x+2y-z-3=0, 3x-y+2z-1=0,$

$2x-2y+3z-2=0, x-y+z+1=0$ Ans. $7x-7y+8z+3=0$

(b) $7x-4y+7z+16=0, 4x+3y-2z+3=0$

$x-3y+4z+6=0, x-y+z+1=0$ Ans. $3x-7y-9z+13=0$

5. Find the smallest angle between the planes

(a) $3x+2y-5z-4=0, 2x-3y+5z-8=0$ Ans. $\cos\theta=25/38$

(b) $3x-4y+5z=0, 2x-y-2z=0$ Ans. $\pi/2$

(c) $2x+2y-7=0, x+2y-z+16=0$ Ans. $30°$

Ex. 13 Find the distance of A (1, − 2, 3) from the line PO through P(2, −3, 5), which makes equal angles with the axes.

The direction cosines of PQ are $1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}$

If p be the required perpendicular distance, then

$p^2 = \{(-3+2)/\sqrt{3} - (5-3)/\sqrt{3}\}^2 + \{5-3)/\sqrt{3} - (2-1)/\sqrt{3}\}^2 + \{(2-1)/\sqrt{3}) - (3+2)/\sqrt{3}\}^2$

$= 1/3, 14$ or $p = \sqrt{(14/3)}$ (See Art. 33).

14. Define the direction cosine of straight line. Find the acute angle between the lines whose direction cosines are given by the relations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$ [1996, 2005]

Solution : Direction cosine : A line makes an angles α β γ with the positive direction of x, y and z axes. Then cos α, cos β, cos γ called direction cosine. It is denoted by l, m, n.

2nd part : We have

$l+m+n=0$ ... ... (1)

$l^2+m^2-n^2=0$ ... ... (2)

From (1) $n = -(l + m)$

Putting the values of n is (2) We get

$l^2 = m^2 - (l + m)^2 = 0$

$\Rightarrow l^2 + m^2 - l^2 - 2lm - m^2 = 0$

$\Rightarrow -2lm = 0$ ... ... (3)

$\Rightarrow l = 0$ ... ... (4)

or, $m = 0$ ...

From (1) and (3)

$l+m+n=0$

$l+om+on=0$

$\therefore \frac{l}{0-0}=\frac{m}{1-0}=\frac{n}{0-1}$

$\therefore \frac{l}{0}=\frac{m}{1}=\frac{n}{-1}=\frac{1}{\sqrt{o^2+1^2+1^2}}=\frac{1}{\sqrt{2}}$

$\therefore l=0, m=\frac{1}{\sqrt{2}}, n=-\frac{1}{\sqrt{2}}$

Again, from (1) and (4) We get,

$l+m+n=0$

$ol+m+on=0$

$\therefore \frac{l}{0-1}=\frac{m}{0-0}=\frac{n}{1-0}$

$\Rightarrow \frac{l}{-1}=\frac{m}{0}=\frac{n}{1}=\frac{1}{\sqrt{1^2+0^2+1^1}}=\frac{1}{\sqrt{2}}$

$\therefore l=\frac{-1}{\sqrt{2}}, m=0, n=\frac{1}{\sqrt{2}}$

Let θ be the angle bet^n them

$\therefore \text{Cos}\theta = \text{Cos}\theta = \pm (l_1l_2 + m_1m_1 + n_1n_2)$

$\therefore \text{Cos}\theta = \pm\left(0.\frac{-1}{\sqrt{2}} + \frac{1}{\sqrt{2}}.0 + \frac{1}{\sqrt{2}}.\frac{1}{\sqrt{2}}\right) = \pm\frac{1}{2} = \cos \pi/3 \text{ or, } \cos 2\pi/3$

$\therefore \theta = \pi/3 \text{ or, } 2\overline{\Lambda}/3$

15. The direction cosines of two st-lines are given by the relations $al + bm + cn = 0$. and $ul^2 + vm^2 + wn^2 = 0$. Prove that the st. lines will be Perpendicular if $a^2(v + w) + b^2(w + u) + c^2(u + v) = 0$ and parallel of $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$ [N.U.H 1997]

Solution : we have

$al + bm + cn = 0$ (1)

$ul^2 + vm^2 + wn^2 = 0$ ... ... (2)

From (i) $l = -\frac{bm + cn}{a}$

Putting the value of l in (2) we get $u\left\{-\frac{bm + cu}{a}\right\}^2 + vm^2 + wn^2 = 0$

$\Rightarrow u(b^2m^2 + 2bcmn + c^2n^2) + a^2vm^2 + a^2wn^2 = 0$

$\Rightarrow (b^2u + a^2v)m^2 + 2bcumn + (c^2u + a^2w)n^2 = 0$

$\Rightarrow (b^2u + a^2v)\frac{m^2}{n^2} + 2bcu\frac{m}{n} + c^2u + a^2w = 0$ ... ... (3)

Which is quadratic eq^n.of $\frac{m}{n}$

Let $\frac{m_1}{n_1}$ and $\frac{m_2}{n_2}$ be the two roots of ... ... (3)

$\therefore \frac{m_1}{n_1}.\frac{m_2}{n_2} = \frac{c^2u + a^2w}{b^2u + a^2v}$

$\frac{m_1m_2}{c^2u + a^2w} = \frac{n_1n_2}{b^2u + a^2v}$

Similarly, $\frac{l_1l_2}{b^2w + c^2v} = \frac{m_1m_2}{c^2u + a^2w}$

$\therefore \frac{l_1l_2}{b^2w + c^2v} = \frac{m_1m_2}{c^2u + a^2w} = \frac{n_1n_2}{b^2u + a^2v} = k \text{ (say)}$

$\therefore l_1l_2 = k(b^2w + c^2v),\ m_1m_2 = k(c^2u + a^2w)$

$m_1n_2 = k(b^2u + a^2v)$

The two lines are perpendicular of $l_1l_2 + m_1m_2 + n_1n_2 = 0$

$\Rightarrow k(b^2w + c^2v) + k(c^2u + a^2w) + k(b^2u + a^2v) = 0$

$\Rightarrow a^2(w + v) + b^2(w + u) + c^2(u + v) = 0$

wo lines are parallel of the two roots of (3) are equal.

$\therefore (2bcu)^2 - 4(b^2u + a^2v)(c^2u + a^2w) = 0$

$\Rightarrow b^2c^2u^2 + b^2c^2u^2 + a^2b^2uw = a^2c^2uv + a^4vw$

6. Find direction cosines of the normals to the planes and length of the normal.

(a) $x + 2y + 2z - 1 = 0$ Ans. $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}$; $p = \frac{1}{3}$

(b) $9x + 6y - 2z + 7 = 0$ Ans. $-\frac{9}{11}, -\frac{6}{11}, \frac{2}{11}$; $p = \frac{7}{11}$.

7. Find the intercepts made by the plane $2x + y - 2z = 3$ on the co-ordinate axes. What ar the d.c.,s of the plane. Ans. $\frac{3}{2}, 3, -\frac{3}{2}$; d.c.s. $(\frac{2}{3}, \frac{1}{3}, -\frac{3}{2})$

8. A plane contains the pt. $(1, -1, -2)$. Its normal has d.r's $3, -2, -1$. Find the equation of the plane. Ans. $3x - 2y - z = 7$

9. Find the equations to the planes

(a) Through the intersections of the planes $x - 2y + 3z + 4 = 0$ and $2x - 3y + 4z - 7 = 0$ and the point $(1, -1, 1)$ Ans. $9x - 13y + 17z - 39 = 0$

(b) Through $(1, 1, 2)$ and perpendicular to each of the planes $2x - 2y - 4z - 6 = 0$ and $3x + y + 6z - 4 = 0$ Ans. $x + 3y - z - 2 = 0$

(c) through $(1, 2, 2)$ and parallel to $3x + 2y + z = 7$ Ans. $3x + 2y + z = 0$

(d) through $(-1, 3, 2)$ and perpendicular to the two planes $x + 3y + 2z = 5$, $3x + 3y + 2z = 8$ Ans. $2x + 4y + 3z + 8 = 0$

10. Find the equation of the plane which is parallel to the plane

(a) $4x - 4y + 7z - 3 = 0$ and distance 4 units from the point $(3, 1, -2)$. Ans. $4x - 4y + 7z + 38 = 0$, $4x - 4y + 7z - 34 = 0$

(b) $2x - 3y - 6z - 14 = 0$ and distant 5 from the origin. Ans. $2x - 3y - 6z \pm 35 = 0$

11. Find the equation of the plane which passes through the poins $(1, 0, -1)$ and $(2, 1, 3)$ and is perpendicular to the plane $2x + y + z = 1$ Ans. $3x - 7y + z = 2$

(a) Find the equation of the plane perpendicular to each of the $x - 4y + z = 0$ and $3x + 4y + z - 2 = 0$ and at a distance unity from the origin. [N.U.H. 2005] [C. U; D. U. 19 77] Ans. $-4x + y + 8z + 9 = 0$ [D.U. 1982]

12. Plane is drawn through the line of intersection of the planes $x + 2y + 2z = 1$, $x + y - z + 1 = 0$ and is at distances. 1 (one) from the point $(4, -2, 1)$. Prove that there are two such planes and fine their equations. Ans. $3x + 4y + 1 = 0$, $3x + 2y - 6z + 5 = 0$ [R.U. 63]

13. Find the equation of the plane which is perpendicular to the plane $5x + 3y + 6z + 8 = 0$ and which contains the line of intersection of the planes $x + 2y + 3z - 4 = 0$, and $2x + y - z + 5 = 0$ Ans. $51x + 15y - 50z + 173 = 0$ [C.U. '80; C.H. '80]

14. Find the equation of plane that passes through $(2, -3, 1)$ and is normal to the line joining the points $(3, 4, -1)$ and $(2, -1, 5)$ Ans. $x + 5y - 6z + 19 = 0$ [D.U. '82; C.U. '78]

15. Find the equation of the plane through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $4x + 3y + 2z + 1 = 0$ and perpendicular to the plane $x + y + z + 9 = 0$ and show that it is perpendicular to xz plane. Ans. $x - z = 2$, the Co-efficient of y is zero.

$\Rightarrow a^2b^2uw + a^2c^2uv + a^4vw = 0$

$\Rightarrow b^2uw + c^2uv = a^2uw = 0$

$\therefore \frac{a^2}{u} + \frac{b^2}{v} = \frac{c^2}{w} = 0$ (Proved)

16. Direction cosine of two straight lines are connected by the relations $2l + 2m - n = 0$ and $mn + nl + lm = 0$ show that the lines are perpendicular each other.[N.U.H 1988]

Solution : We have

$2l + 2m - n = 0$ ... (1)

and $mn + nl + lm = 0$ ... (2)

From (1); $n = 2l + 2m$

Putting this valus in (2) we get,

(2) $\Rightarrow m(2l + 2m) + l(2l + 2m) + lm = 0$

$\Rightarrow 2lm + 2m^2 + 2l^2 + 2lm + lm = 0$

$\Rightarrow 2l^2 + 5lm + 2m^2 = 0$

$\Rightarrow (2l + m)(2m + l) = 0$

$\therefore 2l + m = 0$ ..... (3) and $l + 2m = 0$ ... (4)

From (1) and (3)

$2l + 2m - n = 0$

$2l + m + 0n = 0$

$\therefore \frac{l}{0+1} = \frac{m}{-2-0} = \frac{n}{2-4}$

$\frac{l}{1} = \frac{m}{-2} = \frac{n}{-2} = 1$

$\frac{l}{-1} = \frac{m}{2} = \frac{n}{2} = \frac{1}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{1}{3}$

$l = \frac{-1}{3}, m = \frac{2}{3} = n, \frac{2}{3}$

$\therefore$ dc's of the 1st line is $\frac{-1}{3}, \frac{2}{3}, \frac{2}{3}$

Again, (1) and (4)

$2l + 2m - n = 0$

$l + 2m + on = 0$

$\therefore \frac{l}{0+2} = \frac{m}{-1-0} = \frac{n}{4-2}$

$\frac{l}{2} = \frac{m}{-1} = \frac{n}{2} = \frac{1}{\sqrt{2^2+1^2+2^2}} = \frac{1}{3}$

$\therefore l = \frac{2}{3}, m = -\frac{1}{3}, n = \frac{2}{3}$

$\therefore$ dc's of the 2nd line is $\frac{2}{3}, \frac{-1}{3}, \frac{2}{3}$

The two lines are perpendicular of

$\boxed{l_1l_2 + m_1m_2 + n_1n_1 = 0}$

Now, $-\frac{1}{3}\cdot\frac{2}{3}+\frac{2}{3}\ \frac{-1}{3}+\frac{2}{3}\cdot\frac{2}{3}=\frac{-2}{9}-\frac{2}{9}+\frac{4}{9}=0$

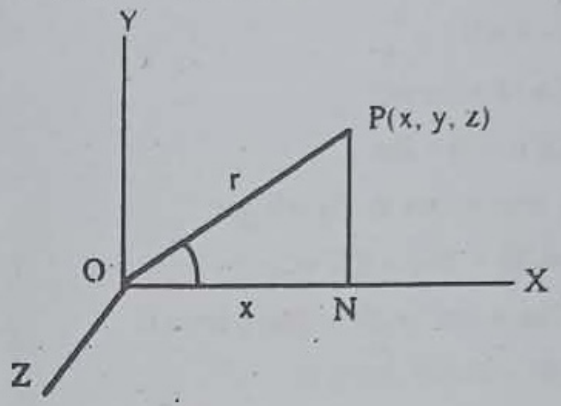
Hence The given lines are perpendicular each other.

17. Prove that $l^2 + m^2 + n^2 = 1$ where l, m, n dc's of any line [(2000) N.U.H 2002, 2008]

Solution : Let P(x, y, z) be the co-ordinates of any point on the line

$\therefore \cos\alpha = \frac{ON}{OP} = \frac{x}{r}$

$\therefore x = r\cos\alpha = rl$



Similarly,

$y = rm$

$z = m$

Also $Op^2 = r^2 = x^2 + y^2 + z^2$

Now, $x^2 + y^2 + z^2 = r^2l^2 + r^2m^2 + r^2n^2$

$\Rightarrow r^2 = r^2(l^2 + m^2 + n^2)$

$\therefore l^2 + m^2 + n^2 = 1$ Proved.

\# **Find the angle between two diagonals of a cube.**

Solution.

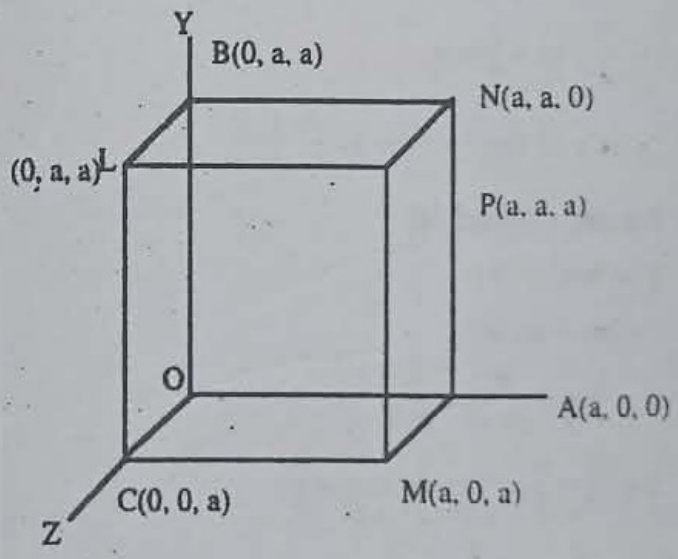
Let OANBPBCM be a cube whose length of each edges a. Let OP and CN be the two diagonals of a cube.

$\therefore$ Co-ordinate of O(0, 0, 0)

\- - - - P (a, a, a)

\- - - - N (a, a, 0)

\- - - - C (0, 0, a)



Dr's of op; a – 0, a – 0, a – 0

ie, a, a, a,

Dr's of CN; a – 0, a – 0, 0 – a

ie, a, a, – a

Let θ be the angle between two diagonals Opened CN,

$\therefore \cos\alpha = \frac{a.a + a.a + a.(-a)}{\sqrt{(a^2+a^2a^2)(a^2+a^2+a^2)}} = \frac{a^2}{\sqrt{3a^2 \times 3a^2}}$

$\cos\theta = \frac{1}{3} \therefore \theta = \cos^{-1}\frac{1}{3}$ (Ans)

(a) $l - 5m + 3n = 0, 7l^2 + 5m^2 - 3n^2 = 0$ Ans. $1/\sqrt{14}, 2/\sqrt{14}, 3/\sqrt{14}; -1/\sqrt{6}, 1/\sqrt{6}, 2/\sqrt{6}$

(b) $l + m - n = 0, nm + 6ln - 12lm = 0$ Ans. $1/\sqrt{26}, 2/\sqrt{26}, 3/\sqrt{26}; 1/\sqrt{14}, 2/\sqrt{14}, 3/\sqrt{14}$

14. If A, B, C, D, be the four points (3, 6, 9), (1, 2, 3), (2, 3, 1) and (4, 6, 2) respectively, prove that the line joining the points A and B intersects the line joining other two points. Find the co-ordinates of the point and the ratios in which this point of intersection divides the two lines.

Ans. -3 " 1 and -1 " 2

15. Prove by the method of projection that if A, B, C, D be four points whose co-ordinates are $(6, -6, 0)$, $(-1, -7, 6)$, $(3, -4, 4)$ and $(2, -9, 2)$ respectively, then AB is perpendicular to CD.

16. Show that $(4, 3, -1)$, $(-1, 4, 7)$, $(\frac{3}{2}, \frac{7}{7}, 3)$ are three collinear points.

17. Show that A (3, 6, 9), B (1, 2, 3); C (2, 3, 1), D (4, 6, 2) are four points such that AB and CD intersect.

18. Show that the lines from A (5, 2, -3) to B (6, 1, 4) and from C (-3, 2, -1) to D (-1, -4, 13) are parallel.

19. Show that the lines AB and BC are perpendicular to each other, where A (-11, 8, 4), B $(-1, -7, -1)$, C (9, -2, 4)

20. Find the direction cosines of the line which is perpendicular to the lines with direction cosines proportional to $(1, -2, -2)$, $(0, 2, 1)$. Ans. 2/3, $-1/3$, 2/3.

21. If $l_1, m_1, n_1, l_2, m_2, n_2$, are the direction cosines of two mutually perpendicular lines, show that the cosines of the line perpendicular to them both are $m_1n_2 - m_2n_1$, $n_1l_2 - n_2l_1$, $l_1m_2 - l_2m_1$

22. Prove that the straight lines whose direction cosines are given by the relations $al + bm + cn = 0$ and $fmn + gnl + hlm = 0$ are perpendicular if $f/a + g/b + h/c = 0$ and parallel if $\sqrt{(af)} \pm \sqrt{(bg)} \pm \sqrt{(ch)} = 0$ [C. U. 1986]

23. Prove that two lines whose direction are cosines are conneceted by the two relations $al - bm + cn = 0$ and $ul^2 + vm^2 + wn^2 = 0$ are perpendicular if

$a^2(v + w) + b^2(w + u) + c^2(u + v) = 0$ and parallel if $\frac{a^2}{u} + \frac{b^2}{u} + \frac{c^2}{w}$ [C. U. 1976, 1980]

24. Prove that the angle between two diagonals of a cube is $\cos^{-1}, \frac{1}{3}$

25. Show that the equation to the right circular cone whose vertex is at the origin whose axis has direction cosines $\cos\alpha$, $\cos\beta$, $\cos\gamma$ and whose semi-vertical angle θ, is

$(y\cos\gamma - z\cos\beta)^2 + (z\cos\alpha - x\cos\gamma)^2 + (x\cos\beta - y\cos\alpha)^2 = (x^2 + y^2 + z^2)\sin^2\theta$

26. l_1, m_1, z_1 ; l_2, m_2, n_2 are the direction cosines of two concurrent lines: show that the direction cosines of two lines bisecting the angle between them are proportional to $(l_1 \pm l_2, m_1 \pm m_2, n_1 \pm n_2)$ [D.U. 1984]

EXERCISE–1

1. Find the distance between the origin and point given.

(a) $(3, 4, -5)$ Ans. $5\sqrt{2}$

(b) $(-4, 0, -2)$ Ans. $2\sqrt{5}$

2. Find the distance between each of the following pairs of points.

(a) $(5, -2, 3), (-4, 3, 7)$ Ans. $\sqrt{(122)}$

(b) $(0, 3, 0), (6, 0, 2)$ Ans. 7

3. Find the co-ordinates of the point which divides the line joining the points $(2, -4, 3)$, $(-4, 5, -6)$ in the ratio $1 : -4$ Ans. $(4, -7, 6)$

4. Find the ratio in which the line joining the points $(2, 4, 5)$ $(3, 5, -4)$ is divided by the YZ plane. Ans. $-2 : 3$, $(0, 2, 23)$

5. If P and Q are $(2, 3, -6)$ and $(3, -4, 5)$ respectively and O be the origin, find the direction cosines of OP, OQ, PQ [R.U. 1980]

Ans. $2/7$ $3/7$. $-6/7$; $3/5\sqrt{2}$, $-4/5\sqrt{2}$, $1/\sqrt{2}$. $(-1, 7, -11)\sqrt{171}$

6. Find the angle between the lines whose direction ratios are (3, 1, 2) and (2, −2. 4).Ans. $\cos^{-1}\sqrt{3}/7$

7. Find the direction cosines of the positive z-axis. Ans. 0, 0, 1

8. Find the cosine of the angle between two directed lines having the dirction cosines

(a) $4/9, -8/9, -1/9$; $-1/3, -2/3, -2/3$. Ans. $\cos^{-1}\sqrt{14/27}$

(b) $1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3}$; $1/2, \sqrt{(6/4)}, -1/2$ Ans. $\cos^{-1} 1/4$

9. Show that the lines whole direction cosines are propotional to 2, 1, 1; 4, $\sqrt{3}-1$, $-\sqrt{3}-1$; $4, -\sqrt{3}-1, \sqrt{3}-1$, are inclined to the another at angle $\pi/3$.

10. If P, Q. are (x_1, y_1, z_1), (x_2, y_2, z_2), the projection of PQ on a line whose direction cosines are $\cos\alpha, \cos\beta, \cos\gamma$ is gven by $(x_2 - x_1)\cos\alpha + (y_2 - y_1)\cos\beta + (z_2 - z_1)\cos\gamma$

11. Find the distance of (−1, 2, 5) from the line through (3, 4, 5) whose direction cosines are proportional to 2, −3, 6. Ans. $4\sqrt{(61/7)}$

12. (a) Find the distance of (−2 , 3, 4) from the line through the point, (− 1, 3, 2) whose direction cosines are proportional to 12, 3, −4. Ans. $\sqrt{445}/13$ [D.U. 1966]

12. (b) The direction cosines l, m, n of two lines are connected by the relations $l + m + n = 0$, $2lm + 2ln - mn = 0$, Find them.

Ans. $1/\sqrt{6}, 1/\sqrt{6}, -2/\sqrt{6}$; $1/6, -2/\sqrt{6}, 1/\sqrt{6}$.

13. The direction cosines of two lines are determined by the relations